FinTrust Digital Bank — Week 2 Project Summary

AnalystLab Africa Experience Lab · Data Analytics Track · 11 required sections (brief §11)

Requirement	Response
1. What you planned to accomplish in Week 2	Per the Week 1 plan: clean and prepare the data, build the analytical SQL pack, perform exploratory analysis in Python, build the management dashboard, and produce documented business findings.
2. What you actually completed	Cleaned both datasets (12,000 and 1,500 rows retained); an 8-sheet Excel Data Quality & Cleaning Workbook; a 10-question SQL analysis file with business interpretations; a Python EDA notebook with 12 explained visualisations; a Power BI dashboard build package; 7 business findings; 14 engineered features; two trained classification models evaluated across 6 metrics plus 5-fold cross-validation.
3. Tools used	Microsoft Excel, SQL (SQLite/DuckDB syntax), Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn), Jupyter notebook, Power BI build package, Git/GitHub.
4. Key findings / development outcomes	Transfer and Cash Withdrawal hold 58% of flagged transactions on 41% of volume; international transactions are reviewed at 1.95x the domestic rate (36.88% vs 18.88%); night transactions ~28% vs ~17% (1.70x); top-decile amounts 2.02x; channel spread is only 4 points, so channel is a weak differentiator; the highest-value customer is not in the Premium segment; mean transaction value (NGN 46,706) overstates the median (NGN 10,306) by ~4.5x.
5. Major challenges encountered	Reconciling the Week 1 assumption that cross-track collaboration begins in Week 2 with the Week 2 brief, which explicitly states it does not. Also: the Location-vs-City trap, the heavy amount skew, and the impossibility of authoring a native Power BI .pbix programmatically.
6. Important decisions made and why	Dropped cross-track exchange (brief policy); imputed missing categoricals as 'Unknown' instead of dropping 96 records (preserve value integrity); kept Location and City separate (12.7% agreement); retained amount outliers (genuine values); added the median to all amount KPIs (skew 3.2); treated every risk figure as synthetic (brief disclaimer).
7. Changes to the Week 1 approach	Three: removed cross-track exchange; added a gradient-boosted model alongside logistic regression; demoted accuracy from headline metric after it was shown to equal the do-nothing baseline (0.801 vs 0.804).
8. Testing or evaluation performed	8-step evaluation log in the required Work Completed → Test → Finding → Action → Result format, covering null handling, the Location/City trap, mean-vs-median, model threshold tuning, cross-validation, channel significance, Power BI reproducibility and SQL-vs-Python reconciliation. All 10 SQL results reconcile exactly with the independent Python computation.
9. Key limitations	All data is synthetic; Risk_Review_Flag is an unknown-provenance educational label with no real ground truth; models achieve only ~0.68 ROC-AUC / ~0.33 PR-AUC; a native .pbix is supplied as an importable build package rather than a binary; SQL date functions differ by engine.
10. Remaining work	Power BI desktop build from the supplied package; richer feature engineering and hyper-parameter tuning; a proper API/serving layer; a written data dictionary for the engineered

	columns.
11. Proposed focus for Week 3	Serve the model behind a tested prediction service; deepen the dashboard with drill-through and drill-down; attempt to lift PR-AUC meaningfully (currently 0.33) via class weighting, hyper-parameter search and additional behavioural features; and convert the 7 findings into prioritised business recommendations.